

Geography Ch-9 (Advanced) — Lakes of India & Wetland Conservation

Advanced / India-applied | GS-I & GS-III (Environment) | Source: Majid Husain + D.R. Khullar
Indian lakes by origin: glacial (Kashmir), tectonic (Wular), lagoon (Chilika/Pulicat), salt (Sambhar)
Extra CA: Loktak phumdis + India's Ramsar network (distinct from Must-Do's Chilika CA)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — Lakes of India by Origin (Majid / Khullar)

■ NEWS TRIGGER

India's lakes map its geomorphology — glacial lakes in the Kashmir Himalaya, the tectonic Wular, the lagoons of the east coast, and salt lakes of arid Rajasthan.

■ INTRO & ORIGIN

Majid notes India's **glacial (Pleistocene) lakes** are confined to the Himalaya — the Peninsula has 'no evidence of Pleistocene glaciation' — while other lakes reflect tectonic, coastal and arid processes.

Himalayan glacial lakes (Gangabal, Sheshnag, Kailash-Kund, Sanasar) sit in ice-carved basins; the Peninsula instead has lagoons, tanks, ox-bows and salt lakes.

■ KEY DATA

Lake / group	Origin	Location
Gangabal, Sheshnag, Sanasar	Glacial	Kashmir Himalaya (Majid)
Sat Tal (7 lakes), Naini	Glacial/tectonic	Kumaun, Uttarakhand
Wular	Tectonic (freshwater)	J&K; — among Asia's largest freshwater lakes
Chilika, Pulicat	Coastal lagoon	Odisha; AP/TN
Sambhar	Salt (inland drainage)	Rajasthan (arid)
Sasthamkotta	Freshwater	Kerala (largest freshwater lake of Kerala)

■ STATIC THEORY

- Majid: **glacial lakes are Himalayan** (Gangabal, Sheshnag, Kailash-Kund, Sanasar) — the **Peninsula shows no Pleistocene glaciation**, so it lacks tarns.
- **Sat Tal** (group of seven lakes near Bhimtal, ~1,370 m) in the Kumaun Himalaya is a migratory-bird haven (Majid).
- **Sasthamkotta** (Kollam, Kerala) is a large freshwater lake and tourist site; **Sambhar** (Rajasthan) is a major inland **salt lake** of an arid drainage basin.
- East-coast **lagoons** (Chilika, Pulicat) form behind sand bars/spits; **Wular** (J&K) is a large tectonic freshwater lake.
- Many Indian lakes double as **Ramsar wetlands**, linking geomorphology to conservation policy.

■ MUST-KNOW FACTS

1. Indian glacial lakes are Himalayan only (Peninsula: no Pleistocene glaciation).
2. Wular (J&K) = tectonic, among Asia's largest freshwater lakes.
3. Sambhar (Rajasthan) = major inland salt lake.
4. Sasthamkotta = Kerala's largest freshwater lake.
5. Chilika & Pulicat = east-coast lagoons.

■ UPSC TRAPS

- ✗ Peninsular India has glacial tarn lakes.
- ✓ The Peninsula shows NO Pleistocene glaciation (Majid) — glacial lakes are Himalayan only.
- ✗ Sambhar is a freshwater lake.
- ✓ Sambhar is a SALT lake in an arid inland-drainage basin (Rajasthan).
- ✗ Pulicat is on the west coast.

✓ Pulicat is an EAST-coast lagoon (Andhra Pradesh/Tamil Nadu).

■ MAINS ANGLE

Map India's lakes onto their formative processes (glacial, tectonic, lagoonal, salt) and link this to their ecological roles and conservation needs.

■ STUDY LINK

Geography → India → Lakes (Majid: Lakes of India; Khullar: Physiography)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Loktak Phumdis & India's Ramsar Network

GS-III · Indian lakes applied → Wetland policy

■ NEWS TRIGGER

Loktak Lake (Manipur) — famous for floating 'phumdis' and the only floating national park (Keibul Lamjao, home of the Sangai) — is on the Montreux Record; India now leads Asia with 85+ Ramsar sites.

■ INTRO & ORIGIN

Loktak shows a lake's ecology as a living system: floating phumdis are matted vegetation/soil that host a unique deer, but the lake is stressed by the Ithai barrage and pollution.

■ STATIC THEORY

- **Phumdis** = floating masses of vegetation, soil and organic matter on Loktak; the **Keibul Lamjao** floating national park sits on them and shelters the endangered **Sangai** deer.
- Loktak is on the **Montreux Record** (Ramsar sites needing priority conservation) due to shrinking phumdis, the Ithai barrage and pollution.
- India has the **highest number of Ramsar sites in Asia (85+)**, spanning lagoons, tectonic, high-altitude and man-made lakes.

■ MUST-KNOW FACTS

1. Loktak (Manipur): floating phumdis; Keibul Lamjao = only floating national park.
2. Sangai deer is endemic to Loktak's phumdis.
3. Loktak is on the Montreux Record; India leads Asia in Ramsar sites (85+).

■ UPSC TRAPS

- ✗ Wular Lake is on the Montreux Record.
- ✓ LOKTAK is on the Montreux Record; Wular is not.

■ MAINS ANGLE

Link lake ecology (phumdis) to wetland degradation and India's Ramsar conservation commitments.

■ STUDY LINK

Geography → India Lakes → Wetlands / Ramsar (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI